Status	Finished
Started	Monday, 12 October 2026, 1:31 PM
Completed	Monday, 12 October 2026, 1:36 PM
Duration	4 mins 39 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94 ✗

b. 89 and 99

c. 90 and 99

d. 89 and 94

Question **2**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search

b. Binary search ✓

c. Hash search

d. Sequential search

Question **3**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted

b. The search ends successfully ✓

c. The search continues in the left sublist

d. The search continues in the right sublist

Question **4**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By comparing each element in the list sequentially ✓
c. By using a hash function
d. By sorting the list first

Question **5**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Linear search ✓
c. Both (1) & (3)
d. Hash search

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. In the entire list
c. In the right sublist
d. At the beginning of the list

Question **7**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is the last element in the array
b. When the item is somewhere in the middle of the array ✓
c. When the item is not the array at all
d. Item is the last element in the array or item is not there at all

Question **8**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O(n log n)
c. O🥎
d. O(1) ✓

Question **9**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Compare the target element with the first element in the list
b. Divide the list into two equal parts ✗
c. Compare the target element with the middle element in the list
d. Sort the list

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search requires the list to be sorted.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By starting from the first element
c. By comparing each element sequentially
d. By using a hash function

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is unsorted
d. When the list is sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓
b. When the list is large and sorted
c. When the list is large and unsorted
d. When the list is small and sorted

Question **14**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. 1

Question **15**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O(1)
c. O(n log n)
d. O👎